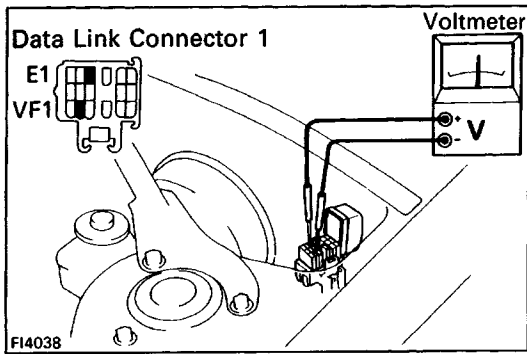




Oxygen Sensor (Main)

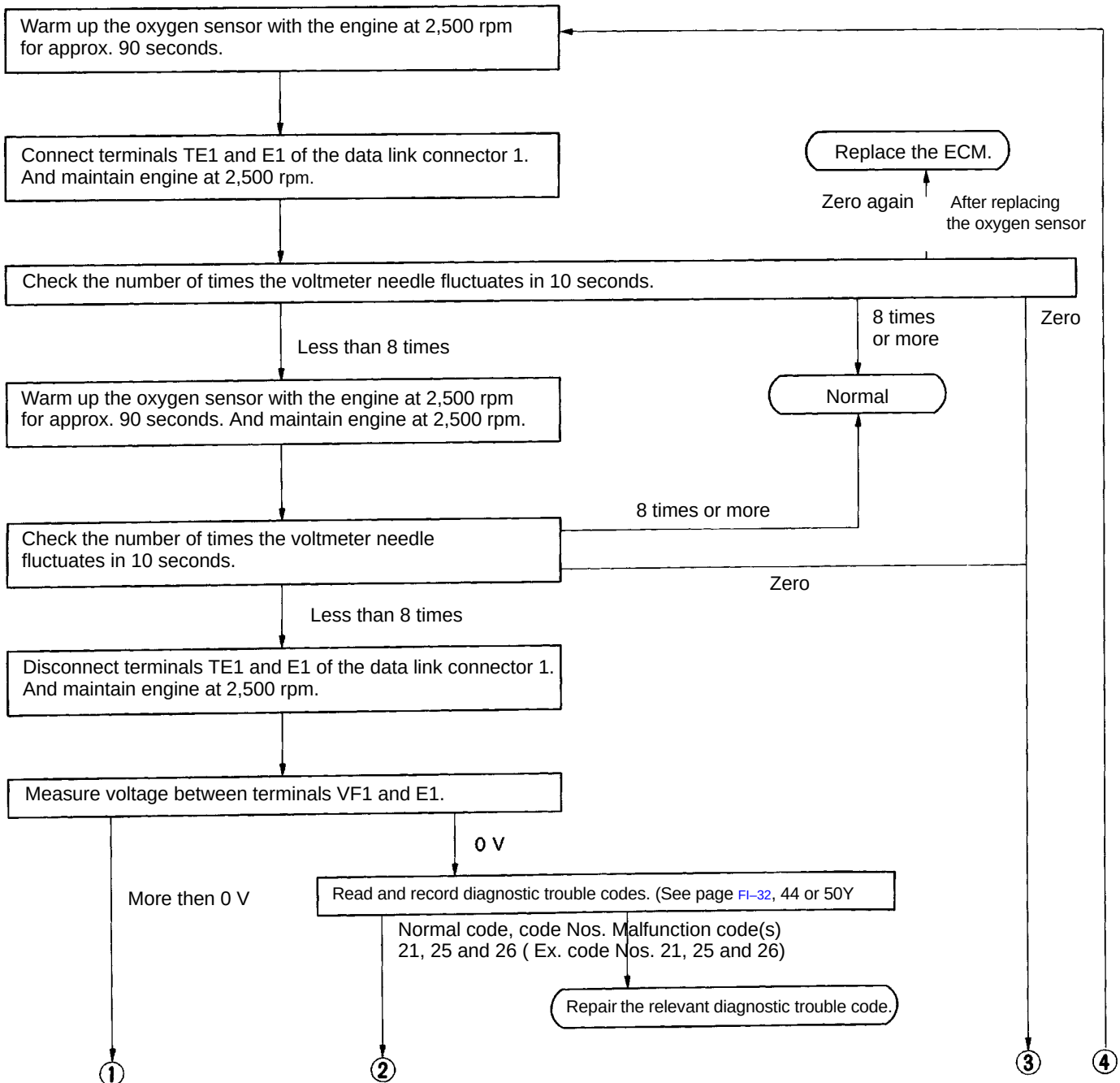
INSPECTION OF OXYGEN SENSOR

1. WARM UP ENGINE

Allow the engine to warm up to normal operating temperature.

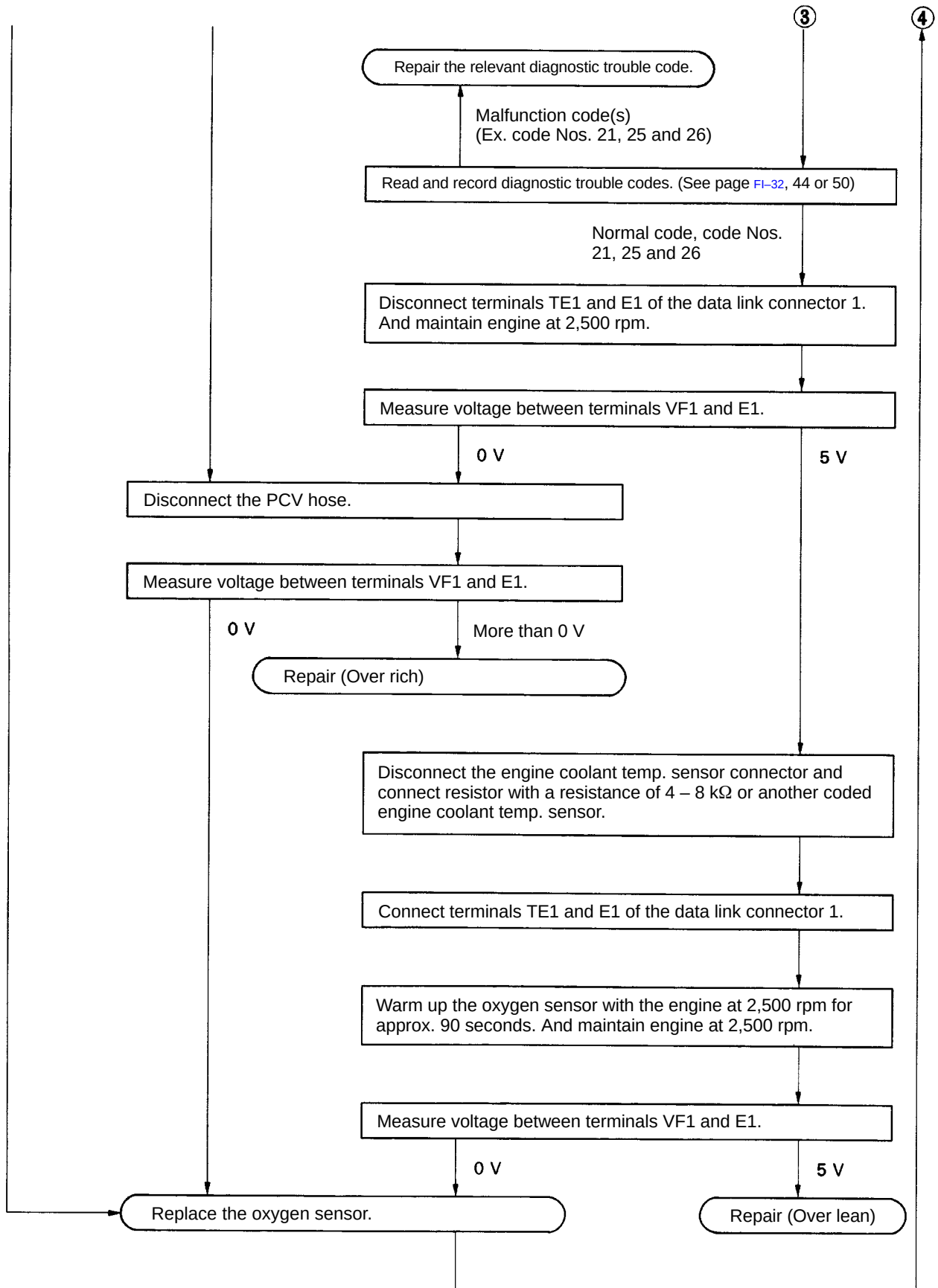
2. INSPECT FEEDBACK VOLTAGE

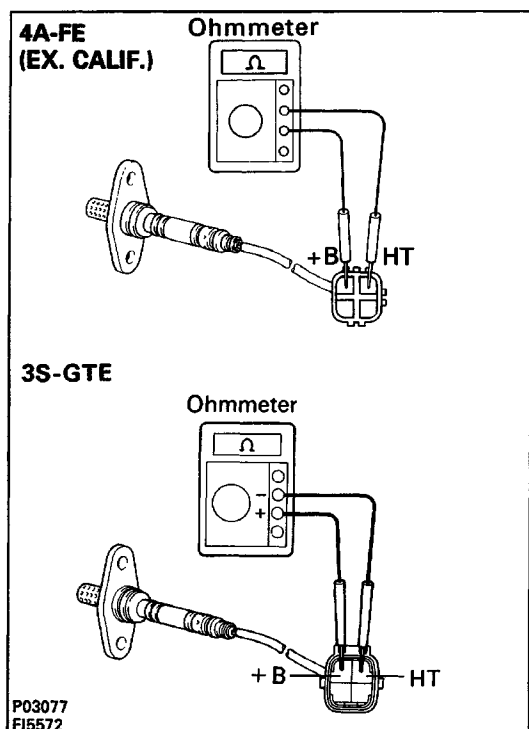
Connect the positive (+) probe of a voltmeter to terminal VF1 of the data link connector, and negative (-) probe to terminal E1. Perform the test as follows:



CONTINUED ON PAGE [FI-238](#)

CONTINUED FROM PAGE FI-237





3. (4A-FE (Ex. CALIF.) AND 3S-GTE)

INSPECT HEATER RESISTANCE OF OXYGEN SENSOR

Using an ohmmeter, measure the resistance between the terminals +B and HT.

Resistance (Cold): 5.1 – 6.3 k at 20°C (68°F)

If the resistance is not as specified, replace the sensor.

Sub-Oxygen Sensor (5S-FE CALIF. only)

INSPECTION OF SUB-OXYGEN SENSOR

INSPECT SUB-OXYGEN SENSOR

HINT: Inspect it only when code No.27 is displayed.

(a) Cancel diagnostic trouble code. (See page [FI-43](#))

(b) Warm up the engine until it reaches normal operating temperature.

(c) (M / T)

Drive for 5 minutes or more between 80 km/h (50 mph) and 100 km/h in 4th or 5th gear.

(A/T)

Drive for 5 minutes or more between 80 km/h (50 mph) and 100 km/h (62 mph) in "D" position.

(d) Following the conditions in step (c), press fully on the accelerator pedal for 2 seconds or more.

HINT: Do not exceed 100 km/h (62 mph), or diagnostic trouble code will be cancelled.

(e) Stop the vehicle and turn the ignition switch OFF.

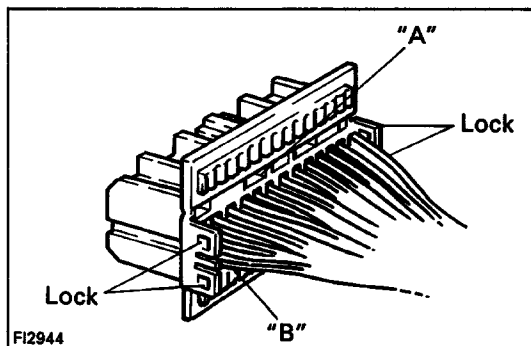
(f) Carry out steps (b), (c) and (d) again to test acceleration.

If code No. 27 appears again, check the sub-oxygen sensor circuit. If the circuit is normal, replace the sub-oxygen sensor.

Electronic Control Module (ECM)

INSPECTION OF ECM

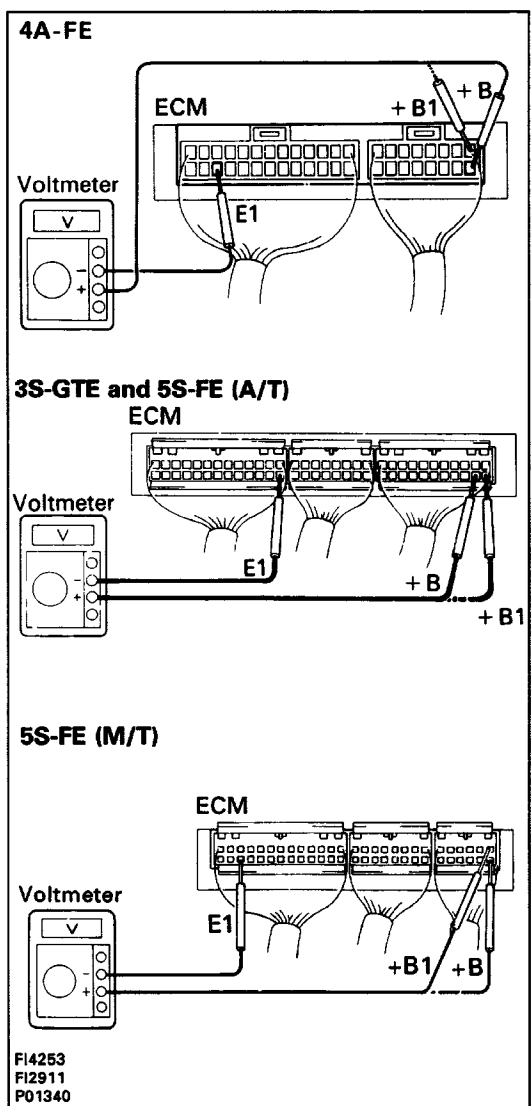
HINT: The MR and SFI circuit can be checked by measuring the resistance and voltage at the wiring connectors of the ECM.



1. PREPARATION

- (a) Disconnect the connectors from the ECM.
- (b) Remove the locks as shown in the illustration so that the tester probe(s) can easily come in.

NOTICE: Pay attention to sections "A" and "B" in the illustration which can be easily broken.



2. INSPECT VOLTAGE OF ECM

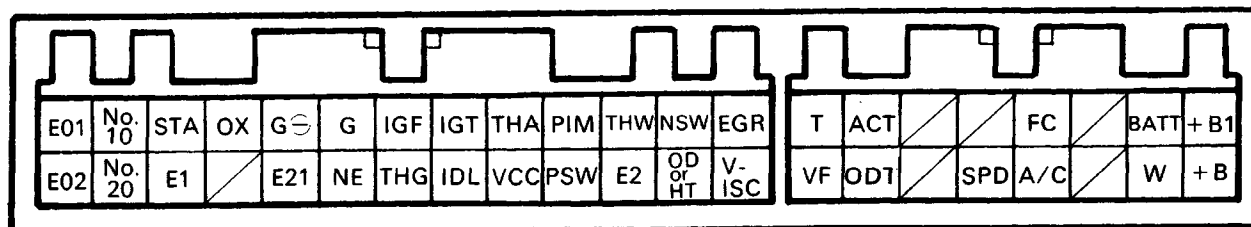
Check the voltage between each terminal of the wiring connectors.

- Turn the ignition switch ON.
 - Measure the voltage at each terminal.
- HINT:
- Perform all voltage measurements with the connectors connected.
 - Verify that the battery voltage is 11 V or more when the ignition switch is ON.

Voltage at ECM Wiring Connectors (4A-FE)

Terminals	Condition		STD voltage (V)
+B + B1 – E1	IG SW ON		10–14
BATT – E 1	–		10–14
I D L – E2	IG SW ON	Throttle valve open	10–14
PSW – E2		Throttle valve fully closed	10–14
PIM–E2	IG SW ON		3.3–3.9
VCC – E2			4.5–5.5
No.10 E01 No.20 E02			10–14
THA – E2	IG SW ON	Intake air temp. 20°C (68°F)	1 –3
THW – E2		Engine coolant temp. 80°C (176°F)	0.1 –1.0
STA – E1	Cranking		6–14
IGT – E1	Cranking or idling		0.7–1.0
W – E1	No trouble (Malfunction indicator– lamp off) and engine running		10–14
A/C – E1	IG SW ON	Air conditioning ON	8–14
ACT – E1		Heater blower SW ON	4–6
T – E1		Data link connector 1 TE1 – E1 not connected	10–14
		Data link connector 1 TE1 – E1 connected	0.5 or less
NSW – E1		Park/Neutral position switch P or N position	0 – 2
		Ex. park/neutral position switch P or N position	6–14
V–ISC – E1	Cranking for ten seconds after starting		10–14

ECM Terminals



Voltage at ECM Wiring Connectors (3S-GTE)

Terminals	Condition		STD voltage (V)
+B – E1 + B1	IG SW ON		10–14
BATT – E 1	–		10–14
I D L – E2	IG SW ON	Throttle valve open	4.5–5.5
VTA – E2		Throttle valve fully closed (Throttle opener must be cancelled first)	0.1 –1.0
		Throttle valve fully open	3.2–4.2
VC – E2		–	4.5–5.5
VS – E2		Measuring plate fully closed	3.7–4.3
		Measuring plate fully open	0.2–0.5
	Idling		1.6–4.1
	3,000 rpm		1.0–2.0
No. 1 No. 2 – E01 No. 3 – E02 No. 4	IG SW ON		10–14
THA – E2	IG SW ON	Intake air temp. 20°C (68°F)	1 –3
THW – E2		Engine coolant temp. 80°C (176°F)	0.1 –1.1
STA – E1	Cranking		6–14
IGT – E1	Cranking or idling		0.8–1.2
RSC – E1 RSO	IG SW ON	ECM connectors disconnected	8–14
W–EI	No trouble (malfunction indicator lamp off) and engine running		10–14
PIM – E2	IG SW ON		2.5–4.5
AC –E1	IG SW ON	Air conditioning ON	8–14
*1 TVIS – E1		Throttle valve fully closed	2.0 or less
		Throttle valve open	10–14
*2 TVIS – E1	Idling		2.0 or less
	4,200 rpm or more		10–14
TE1 – E1	IG SW ON	Data link connector 1 TE1 – E1 not connected	10–14
		Data link connector 1 TE1 – E1 connected	0.5 or less

ECM Terminals

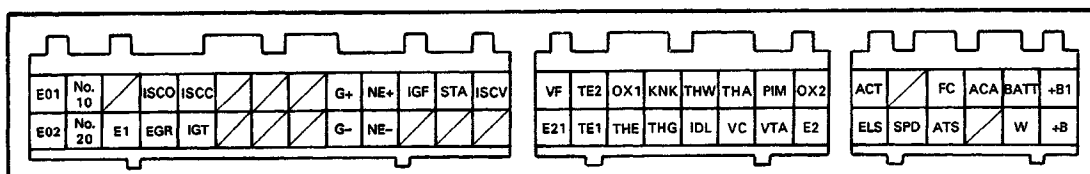
*1 w/ Regular Unleaded Gasoline *2 w/ Premium Unleaded Gasoline

E01	No. 1	No. 2	RSO	RSC	HT	STJ	EGR	G2	NE	IGF	TPC	TVIS	VF	OX	PIM	THW	THA	VS	VC	STA	AC	SPD	ATS	FPR	W	STP	ELS	BATT
E02	No. 3	No. 4				IGT		G1	G $\ominus$			E1	TE1	TE2	KNK	IDL	VTA	THG	E2	ACT						FC	+B1	+B

Voltage at ECM Wiring Connectors (5S-FE)

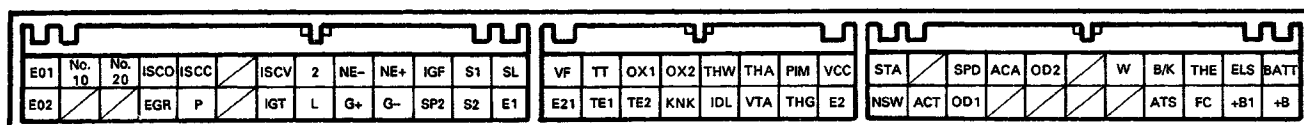
Terminals	Condition		STD voltage (V)
+B + B1 – E1	IG SW ON		10–14
BATT – E1	–		10–14
IDL – E2	IG SW ON	Throttle valve open	8–14
VTA – E2		Throttle valve fully closed (Throttle opener must be cancelled first)	0.8 – 1.2
		Throttle valve fully open	3.2 – 4.2
PIM–E2	IG SW ON		3.3 – 3.9
VC – E2			4.5 – 5.5
No. 10 E01 No. 20 E02			10 – 14
THA – E2	IG SW ON	Intake air temp. 20°C (68°F)	1.9 – 2.9
THW – E2		Engine coolant temp. 80°C (176°F)	0.1 –1.1
STA – E1	Cranking		6 –14
IGT – E1	Cranking or idling		0.8 – 1.2
ISCC – E1 ISCO	IGSWON	ECM connectors disconnected	8 –14
W–E1	No trouble (malfunction indicator lamp off) and engine running		10 – 14
A/C – E1	IG SW ON	Air conditioning ON	8 –14
ACT – E1		Air conditioning ON	4.5 – 5.5
ACA – E1		Air conditioning ON	2 or less
TE1 – E1		Data link connector 1 TE1 – E1 not connected	10–14
		Data link connector 1 TE1 – E1 connected	1 or less
NSW – E1		Park/Neutral position switch P or N position	0 – 2
		Ex. park/neutral position switch P or N position	6 –14
B/ K – E1	Stop light SW ON (Brake pedal depressed)		10 –14

ECM Terminals

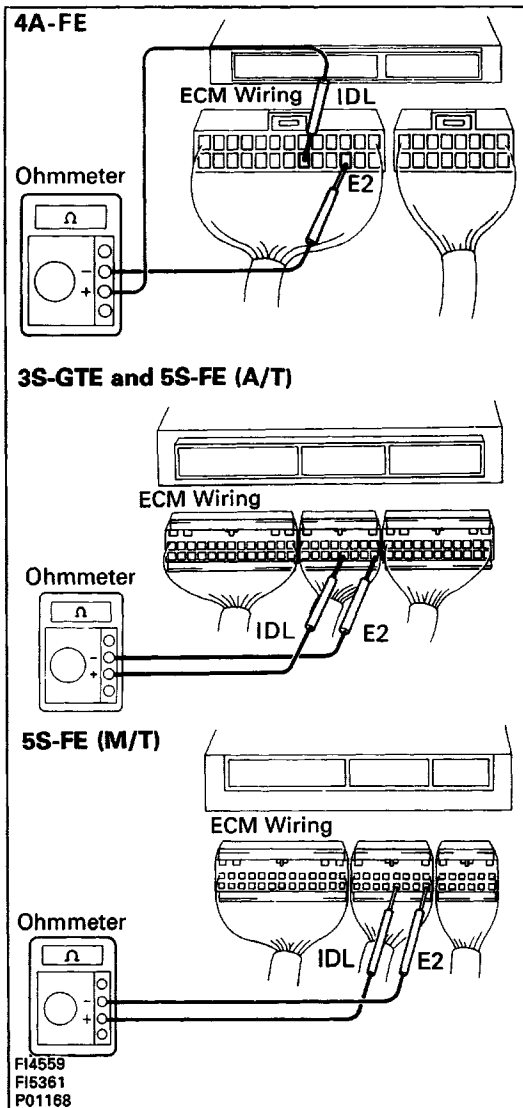


P01821

ECM Terminals



FI2796



2. INSPECT RESISTANCE OF ECM

NOTICE:

- Do not touch the ECM terminals.
- The tester probe should be inserted into the wiring connector from the wiring side.

Check the resistance between each terminal of the wiring connectors.

- Disconnect the connectors from the ECM.
- Measure the resistance at each terminal.

Resistance of ECM Wiring Connector (4A-FE)

Terminals	Condition	STD resistance (Ω)
IDL – E2	Throttle valve open	Infinity
	Throttle valve fully closed	0
PSW – E2	Throttle valve fully open	0
	Throttle valve fully closed	Infinity
THA – E2	Intake air temperature 20°C (68°F)	2,000 – 3,000
THW – E2	Engine coolant temperature 80°C (176°F)	200–400
G1 NE – G (–)	Cold	185–265

ECM Terminals

E01	No. 10	STA	OX	G ⊖	G	IGF	IGT	THA	PIM	THW	NSW	EGR	T	ACT	/	/	FC	/	BATT + B1	
E02	No. 20	E1	/	E21	NE	THG	IDL	VCC	PSW	E2	OD or HT	V- ISC	VF	ODT	/	/	SPD	A/C	/	W + B

FI4266

Resistance of ECM Wiring Connectors (3S-GTE)

Terminals	Condition	STD resistance (Ω)
IDL – E2	Throttle valve open	Infinity
	Throttle valve fully closed	2,300 or less
VTA – E2	Throttle valve fully open	3,100 – 12,100
	Throttle valve fully closed	470–6,100
VC – E2	–	3,900 – 9,000
VS – E2	Measuring plate fully closed	200–600
	Measuring plate fully open	20–1,200
THA – E2	Intake air temp. 20°C (68°F)	2,000 – 3,000
THW – E2	Engine coolant temp. 80°C (176°F)	200–400
G1 G2 – G(–)	Cold	125–190
NE – G(–)	Cold	155–240
RSC + B RSO + B1	–	19.3–22.3

ECM Terminals

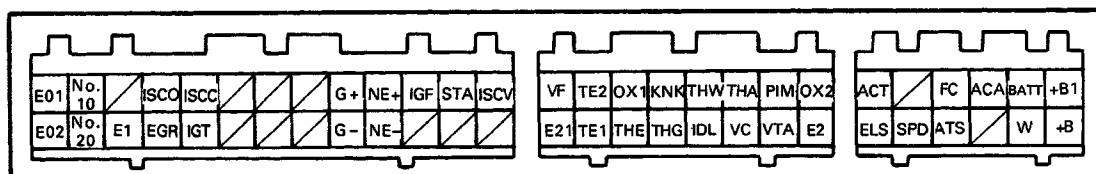
E01	No. 1	No. 2	RSO	RSC	HT	STJ	EGR	G2	NE	IGF	TPC	TVIS	VF	/	OX	PIM	THW	THA	VS	VC	STA	AC	SPD	/	ATS	FPR	W	STP	/	ELS	BATT
E02	No. 3	No. 4	/	/	/	IGT	/	G1	G $\ominus$	/	/	E1	TE1	TE2	KNK	IDL	VTA	THG	E2	/	ACT	/	/	/	/	/	/	/	FC	+B1	+B

FI2796

Resistance of ECM Wiring Connectors (5S-FE)

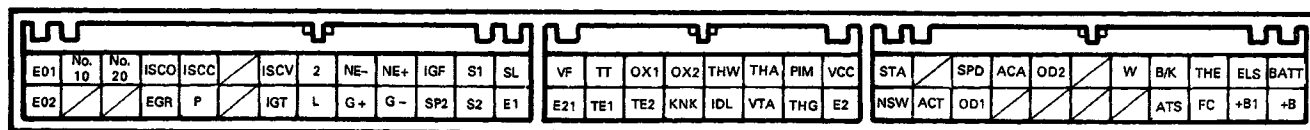
Terminals	Condition	STD resistance (Ω)
IDL – E2	Throttle valve open	Infinity
	Throttle valve fully closed (Throttle opener must be cancelled first)	2,300 or less
VTA – E2	Throttle valve fully open	2,000 – 10,200
	Throttle valve fully closed (Throttle opener must be cancelled first)	200–5,700
VC – E2	—	2,500 – 5,900
THA – E2	Intake air temp. 20°C (68°F)	2,000 – 3,000
THW – E2	Engine coolant temp. 80°C (176°F)	200–400
G+ – G–	Cold	185–265
NE+ – NE–	Cold	370–530
ISCC _ + B ISCO _ +B I	—	19.3–22.3

ECM Terminals



P01821

ECM Terminals



FI2796